Heriot-Watt University

Department of Actuarial Mathematics and Statistics

Life Insurance Mathematics B

Tutorial 10

1. On 1 January 1970 Sam joins an occupational pension scheme at exact age 30. Sam's initial salary rate is £20,000 p.a.. Sam is promoted on 1 January 1980 and 1 July 1997, with a new salary of £30,000 and £40,000 respectively. At age 60 Sam retires and receives a pension of 1/60th of his pensionable salary for every year of service. How much is Sam's pension if pensionable salary is defined in each of the following ways.

   (a) final salary rate;

   (b) average of the salaries earned in the 5 years before retirement

   (c) average of the salaries earned throughout the period of service?

2. In this question, use the service table in 'Formulae and Tables for Actuarial Examinations (2002)' (the Yellow Book) page 142.

   A pension scheme has 1,000 members each aged 50 exactly. Over the last year each member earned £17,000. Salaries increase continuously through time. Calculate the following:

   (a) the expected number of deaths over the next year;

   (b) the expected number of ill-health retirements over the next 3 years;

   (c) the expected salary rate of the surviving members at exact age 51.

3. Discuss the key differences between defined contribution and defined benefit pension schemes - from the point of view of scheme members and employer sponsors. You should consider contributions, benefits and risk in your answer.

Question 4 (taken from a past exam paper)

A pension scheme member is age 50 exactly. She will retire at age 60. Her salary is currently £100,000. The pension scheme calculates Final Pensionable Salary (FPS) as the average of earnings in the last 3 years before retirement.

Calculate the member's FPS in each of the following cases:

(a) The member's salary increases by 3% halfway through each year (ie, at age 50.5, 51.5, etc)

(2 marks)

(b) The member's salary increases countinuously at a rate $\delta = 3\%$ (ie, the member's salary at time t is $100000e^{\delta t}$).

(2 marks)

[Total 4 marks]

Question 5 (OPTIONAL this year)

Mrs Jones joined her employer's career average pension scheme when she was 45 (exact) and she is now aged 65 (exact). Her husband, Mr Jones, is now aged 66 (exact). The pension scheme will pay a pension to Mrs Jones from age 65 until death. If Mr Jones is still alive on Mrs Jones' death, it will pay a pension to Mr Jones until his death.

Calculate the "past service liability" for these pensions immediately before Mrs Jones retires at age 65 exact.

In this calculation, the "past service liability" means the expected present value of the pensions to Mrs Jones and Mr Jones.

Pension scheme details

- *Annual pension paid to Mrs Jones: 1/80 of career average salary for each year of service*
- *Annual pension paid to Mr Jones after death of Mrs Jones: 50% of Mrs Jones' annual pension*
- *All pensions paid annually in advance*
- *No pension increases in payment*

Mrs Jones' earnings history
£60,000 a year at age 45 (exact) increasing by 2% at the end of each year

Basis
Mortality: PFA92 and PMA92
Discount rate: 4%

[Total 4 marks]